SkillNavora Academy · Master Freelancing

AI and Automation

Ten core professions and the AI tools, workflows and projects behind them

AI video · AI integration · Chatbots · Machine learning · AI developer · Automation · n8n · Vibe coding · Cloud · AI consulting

Department 2 of 7 · Draft · Official site: www.skillnavora.in

Tool names, features and prices change. Official links were not checked live in this draft; confirm before publication.

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How to use this textbook

This textbook teaches AI and Automation from absolute zero to freelance-ready. It follows the same structure as the Development and IT textbook, and links to it where skills overlap.

Who it is for

School students (supervised), college students, graduates, working professionals, career changers, freelancers and small business owners.

Safety and responsibility

- Never put personal, customer or medical data into a public AI tool unless its data terms allow it.
- Every AI output is a draft. A human checks facts, numbers and promises before anything reaches a customer.
- Automations that send messages or money need an approval step first.

VERIFICATION NOTE

Tool names, features, plan names and prices change often. The links below point to official sites but were not checked live in this session. Confirm each one before you rely on it or quote it to a client.

PART 1 · MODULE A

AI fundamentals in plain language

The ideas behind every AI tool you will use, explained first in everyday words.

A1. Artificial intelligence

Simple explanation. A computer system that does tasks that normally need human judgement, such as reading text, recognising pictures or answering questions.

A2. Machine learning and deep learning

Machine learning: teaching a computer to find patterns from examples. Like a teacher who shows many photos of cats and dogs until a child can tell them apart. **Deep learning:** machine learning with many layers of simple calculations, used for images, speech and large language models.

A3. Generative AI and large language models (LLMs)

Simple explanation. Generative AI creates new text, images, audio or video. An LLM is a text model trained on large amounts of writing. It predicts the next piece of text, token by token (see the Development and IT textbook for the basics of an API).

A4. Prompt engineering

Writing clear instructions for a model: role, context, task, input, rules and the output format. A weak prompt says 'write a post'. A strong prompt says who the post is for, the length, the tone, the facts that must appear, and the format.

A5. AI agents and agentic AI

Simple explanation. An agent is an AI that can take steps with tools: read an email, look up a sheet, draft a reply and wait for approval. Agentic AI means systems that plan and act over several steps.

A6. Retrieval-augmented generation (RAG)

Simple explanation. Before the AI answers, the system finds the most relevant passages in your own documents and gives them to the model. The answer then cites its sources. Like an open-book exam.

A7. APIs and webhooks

An API lets your software ask an AI model a question and get a reply. A webhook is the reverse: a service calls your address when something happens (a form is submitted).

A8. Traditional automation versus AI automation

Traditional automation	AI automation
Follows fixed rules: if the amount is over 500, send to the manager.	Understands text and decides a category: is this email a complaint, an enquiry or spam?
Predictable and easy to test.	Flexible but can be wrong; needs validation and review.

A9. Hallucinations and model limits

A model can state false facts with confidence. It may not know recent events, your prices or your policies. Fix: give it the source text (RAG), ask it to say 'not found' when unsure, and check every number.

A10. Data privacy and responsible AI

- Know where data goes and whether it is used to train models on your plan.
- Remove personal data before sending text to a model when you can.
- Disclose AI-generated content where your market or platform requires it.
- Keep a human owner for decisions that affect people.

PART 1 · MODULE B

AI tools and platforms: what each one is for

Twenty-five tools across assistants, APIs, automation, agent frameworks, coding and cloud.

Each tool below is listed with what it does, the free and paid options (confirm current plans on the official page), and the official link. For each tool, the learning path follows the pattern: account, first task, a business use case, an API or automation use, and the limits. Pricing is not stated as fact here.

Tool	What it does	Free or paid	Official link (not checked live)	Company
ChatGPT	General assistant for writing, research and analysis	Free tier; paid plans	chatgpt.com	OpenAI
Claude	General assistant; long documents; coding help	Free tier; paid plans	claude.ai	Anthropic
Gemini	General assistant with Google Workspace links	Free tier; paid plans	gemini.google.com	Google
Microsoft Copilot	Assistant inside Microsoft 365 and Windows	Free tier; paid plans for Microsoft 365	copilot.microsoft.com	Microsoft
Perplexity	Search-style answers with sources	Free tier; paid plans	www.perplexity.ai	Perplexity
OpenAI API	Build your own app on OpenAI models	Pay per use	platform.openai.com	OpenAI
Anthropic API	Build your own app on Claude models	Pay per use	console.anthropic.com	Anthropic
Google AI Studio	Try and build with Gemini models	Free tier; pay per use	aistudio.google.com	Google
n8n	Workflow automation with AI steps	Free self-hosted; paid cloud	n8n.io	n8n
Make	Visual automation scenarios	Free tier; paid plans	www.make.com	Make
Zapier	Simple automations across thousands of apps	Free tier; paid plans	zapier.com	Zapier
LangChain / LangGraph	Frameworks for LLM apps and agent workflows	Open source; paid platform	www.langchain.com	LangChain
Flowise	Visual builder for LLM flows (open source)	Open source; paid cloud	flowiseai.com	Flowise
Voiceflow	Design and build conversational agents	Free tier; paid plans	www.voiceflow.com	Voiceflow
Botpress	Chatbot builder with a visual editor	Free tier; paid plans	botpress.com	Botpress
Replit	Browser coding environment with an AI agent	Free tier; paid plans	replit.com	Replit
Lovable	Build full-stack apps from a conversation	Free tier; paid plans	lovable.dev	Lovable

Bolt	Prompt-to-app builder	Free tier; paid plans	bolt.new	Stack Blitz
Cursor	AI code editor	Free tier; paid plans	cursor.com	Anysphere
Claude Code	Coding agent in the terminal and IDE	Included with some Claude plans; API pay per use	docs.claude.com/en/docs/claude-code/overview	Anthropic
GitHub Copilot	AI pair programmer in editors	Free tier for some users; paid plans	github.com/features/copilot	GitHub
Vercel	Hosting for web and AI apps	Free hobby tier; paid plans	vercel.com	Vercel
AWS	Cloud services, including AI services	Free tier; pay per use	aws.amazon.com	Amazon
Microsoft Azure	Cloud and AI services	Free account; pay per use	azure.microsoft.com	Microsoft
Google Cloud	Cloud and AI services	Free tier; pay per use	cloud.google.com	Google

How to learn any tool in a week

1. Day 1: create the account, read the official quick start, and complete one task.
2. Day 2: run one business example from your profession.
3. Day 3: test the limits: long input, wrong facts, slow responses.
4. Day 4: compare it with one alternative from the table.
5. Day 5: write a one-page note: when to use it, when not to, and what it costs.

VERIFICATION NOTE

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DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 1

AI video creator and editor

Profession 1 of 10 in Development and IT.

What the profession means

An AI video creator uses generative tools to make short videos from text, images or a script, and an AI video editor combines those clips with voiceovers, subtitles, music and branding into a finished video for a client.

What professionals do every day

Writes scripts, designs storyboards, generates and edits clips, records or generates voice, adds subtitles and branding, exports for each platform and delivers the files with a usage note.

Why businesses hire them, and what problems they solve

Short video is the main format for advertising, product pages and social media. AI tools cut the time from idea to finished video, so businesses can test more ads and explain products more often.

- Ads and explainers that take too long to produce.
- Multilingual versions for new markets (translation and dubbing).
- Consistent branding across many short videos.

Skills you need

- Scripting and storyboarding for short video.
- Prompting image and video generators with control over style and length.
- Voice generation, subtitles and translation.
- Editing: cuts, timing, colour, audio levels.
- Rights: licensed music, fonts, and disclosure of AI-generated content.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Runway	Text and image to video; editing tools	Free credits	Paid plans	runwayml.com
HeyGen	AI avatar videos and translation	Free trial	Paid plans	www.heygen.com
Synthesia	AI presenter videos	Free trial	Paid plans	www.synthesia.io
Pika	Text and image to video	Free tier	Paid plans	pika.art
Kling AI	Text and image to video	Free tier	Paid plans	klingai.com
Descript	Edit video by editing text; transcription	Free tier	Paid plans	www.descript.com
ElevenLabs	Voice generation and dubbing	Free tier	Paid plans	elevenlabs.io

CapCut	Editing app with templates and captions	Free	Paid features	www.capcut.com
Canva	Design and simple video editing	Free	Paid plans	www.canva.com
Adobe Premiere Pro	Professional video editing	Trial	Subscription	www.adobe.com/products/premiere.html

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	30-second videos from a script with one AI video tool and captions.
Intermediate	Ads with voiceover, branding and two platform sizes.
Advanced	A multilingual version with translated subtitles and dubbed voice, with a review step.

Projects by level

- **Beginner:** A product explainer for a fictional shop.
- **Intermediate:** A 15-second social reel series with a consistent look.
- **Advanced:** A multilingual ad campaign package for a client.

Client-ready portfolio project

A complete 30-second advertisement for a fictional business, delivered as a package (see the assignment below), and a three-video portfolio.

Worked example

ASSIGNMENT

30-second ad for a fictional tutoring centre

Business problem: A tutoring centre wants a 30-second ad for social media that explains its promise and drives free-trial bookings.

Steps

1. Select a fictional business (for example a tutoring centre) and write its one-line promise.
2. Identify the audience: who watches, on which platform, and what they need in the first three seconds.
3. Develop one concept in one sentence.
4. Write a 30-second script in a table: time, visual, narration, on-screen text.
5. Create a storyboard of six frames.
6. Generate visuals with an image or video tool; keep a log of prompts and settings.
7. Generate the voiceover with a text-to-speech tool and check pronunciation.
8. Add subtitles (generated, then corrected by you).
9. Edit in a video editor: cut to the beat, add logo and colours.
10. Export in the platform's recommended size and length.
11. Prepare a client delivery package: video, captions file, script, prompt log and usage notes.

Expected output: A 30-second video in two sizes, captions file, script, storyboard, prompt log and a usage note.

SOLUTION (ANSWER KEY)

Sample 30-second script (tutoring centre):

Time	Visual	Narration	On-screen text
0-3 s	Student frowns at a maths sheet	Stuck on algebra?	Stuck on algebra?
3-10 s	Friendly tutor explains with a diagram	Our tutors break every topic into small steps.	Small steps, clear results
10-20 s	Student solves a problem and smiles	Weekly tests show the progress.	Weekly progress tests
20-30 s	Logo and booking link	Book a free trial class today.	Free trial class

Quality checklist: one clear promise; text readable on a phone; subtitles checked; no claims you cannot prove (for example 'guaranteed marks'); licensed music and fonts; the logo and colours match the brand; file exported at the platform's size.

Assignment with full solution

['Using AI faces or voices of real people without written permission.', 'Skipping the subtitle check; generated captions often mishear names.', 'Making claims in the video that the client cannot prove.',

'Exporting in one size and cropping badly on other platforms.']

Common mistakes

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Freelancing opportunities

Worked example. A tutoring-centre ad: a three-second hook (the problem), a middle that shows the solution, and a call to action with a single offer. Each scene is written in the script table before any tool is opened.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- CapCut resources
- ElevenLabs documentation
- Runway learning resources

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 2

AI integration developer

Profession 2 of 10 in Development and IT.

What the profession means

An AI integration developer connects AI models to the software a business already uses: the website, the CRM, the help desk, the spreadsheet. The work is about data flow, reliability, cost and safety, not about training models.

What professionals do every day

Reads the business process, chooses the model and tool, writes the API calls, handles errors and rate limits, validates outputs, logs decisions and monitors cost.

Why businesses hire them, and what problems they solve

Businesses have data in many systems. An integration makes an AI step part of the process, such as classifying enquiries or drafting replies, without changing the whole system.

- Repetitive reading and sorting of enquiries.
- Summaries and drafts that staff must currently write from scratch.
- Data scattered across tools.

Skills you need

- APIs, JSON and webhooks (Development and IT textbook, Part 1).
- Prompting and structured output (JSON with a schema).
- Error handling, retries and rate limits.
- Security: keys, personal data, logs.
- Cost control: tokens, caching, model choice.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
OpenAI API	Models for text, classification and tools	Pay per use	Pay per use	platform.openai.com/docs
Anthropic API	Claude models for text and tools	Pay per use	Pay per use	docs.claude.com
Google AI Studio	Try and build with Gemini	Free tier	Pay per use	aistudio.google.com
n8n	Connect apps with AI steps	Self-hosted free	Cloud plans	n8n.io
Postman	Test API calls	Free	Paid teams	www.postman.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	A script that sends one text to a model and gets JSON back.
Intermediate	A workflow that classifies form submissions and writes them to a sheet.
Advanced	A CRM integration with validation, logging and a human review queue.

Projects by level

- **Beginner:** Enquiry classifier for a clinic.
- **Intermediate:** Draft-reply assistant with approval.
- **Advanced:** Multi-system pipeline with monitoring and cost report.

Client-ready portfolio project

An AI-assisted enquiry pipeline for a fictional business: form to model to CRM, with validation, approval for sends, logging and a cost estimate.

Worked example

ASSIGNMENT

Classify and validate enquiries

Business problem: A clinic receives enquiries by web form. Staff want the enquiry sorted into booking, price or complaint, with the wrong answers caught before they reach anyone.

Steps

1. Write a prompt that returns only JSON with one field, category, and the three allowed values.
2. Call the model (see the example) and parse the JSON.
3. Reject any value outside the allowed list and send that enquiry to a human.
4. Log the input id, category and time. Do not log personal details.
5. Test with ten sample enquiries, including one that is ambiguous.

Expected output: A function that returns one of three categories, rejects others, and logs safely.

SOLUTION (ANSWER KEY)

The prompt asks for JSON only; the code validates it, because models sometimes add text or invent values. Ambiguous enquiries are sent to a human. The log holds an id and the category, never the name or phone number. Test the rejection path as carefully as the success path.

PYTHON

```
ALLOWED = {"booking", "price", "complaint"}
def classify(enquiry, call_model):
    raw = call_model(f"Return only JSON {{\"category\": one of booking|price|complaint}} for:
{enquiry}")
    try:
        category = json.loads(raw)["category"]
    except (ValueError, KeyError):
```

```
        return "human_review"
    return category if category in ALLOWED else "human_review"

# offline test with a fake model
import json
assert classify("Can I book Monday?", lambda p: '{"category": "booking"}') == "booking"
assert classify("Refund please", lambda p: "Sure, it's a complaint!") == "human_review"
assert classify("Hi", lambda p: '{"category": "spam"}') == "human_review"
print("integration tests passed")
```

Assignment with full solution

["Trusting the model's output without validation.", 'Logging personal data.', 'Sending messages without a review step.', 'Ignoring cost: long inputs multiply the bill.']

Common mistakes

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Freelancing opportunities

Worked example. A form submission is sent to a model with an instruction to return `{"category": "..."}`. Your code checks the category is one of four allowed values; anything else goes to a human. The rule lives in code, not only in the prompt.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- OpenAI API documentation
- Claude documentation
- Google AI for developers

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 3

Chatbot developer

Profession 3 of 10 in Development and IT.

What the profession means

A chatbot developer designs and builds conversations for websites, WhatsApp and messaging apps. Simple bots follow scripted intents; advanced bots use an AI model and the business's own documents, with a clear handover to a person.

What professionals do every day

Maps the conversations customers have, writes intents and answers, builds the bot in a builder or code, connects it to a CRM or calendar, tests with real questions and monitors the transcripts.

Why businesses hire them, and what problems they solve

Customers ask the same questions at any hour. A bot answers the common ones instantly and passes the rest to staff with the context already captured.

- Missed enquiries outside office hours.
- Staff answering the same questions again and again.
- Leads that go cold because the reply is slow.

Skills you need

- Conversation design: intents, slots, fallbacks and handover.
- Python or JavaScript for custom logic.
- Webhooks and APIs (Development and IT, Part 1).
- RAG for answers from documents (see Module D).
- Privacy, consent and opt-out for messaging channels.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Botpress	Visual chatbot builder	Free tier	Paid plans	botpress.com
Voiceflow	Design conversational flows	Free tier	Paid plans	www.voiceflow.com
Flowise	Open-source LLM flow builder	Open source	Paid cloud	flowiseai.com
WhatsApp Business Platform	Messaging channel for businesses	Test tools	Per-message pricing (check current)	developers.facebook.com/docs/whatsapp
Python	Scripts and logic	Free	Free	www.python.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	A rule-based FAQ bot with three intents and a handover.
Intermediate	A WhatsApp lead bot with a template reply and a calendar link.
Advanced	A knowledge assistant that answers from your documents with sources.

Projects by level

- **Beginner:** Website FAQ bot for a clinic.
- **Intermediate:** Lead-capture bot connected to a CRM.
- **Advanced:** Knowledge assistant over a product manual, with a monthly test set.

Client-ready portfolio project

Three chatbot projects (see Module D): a FAQ bot, a lead bot and a document assistant, each with a test log and a handover path.

Worked example

ASSIGNMENT

A FAQ bot with a human handover

Business problem: A coaching centre wants a website chat that answers price, timing and booking questions, and hands everything else to a counsellor.

Steps

1. Write four intents (greeting, price, timing, booking) with keywords.
2. Add a fallback that hands over to a person.
3. Test with ten messages, including words that contain a keyword by accident (for example 'this' contains 'hi').
4. Record each unmatched message for review.

Expected output: A bot that passes the test set and a list of unmatched messages.

SOLUTION (ANSWER KEY)

Matching whole words avoids false matches such as 'hi' inside 'this'. Anything unmatched is handed over, so no question is silently ignored. The unmatched list is the best source for improving the bot.

PYTHON

```
import re

INTENTS = [
    ("greeting", ["hi", "hello", "hey"], "Hello! I can help with prices, timings and
bookings."),
    ("price", ["price", "cost", "fee", "fees", "how much"], "Our fees start at Rs 5,000. Tell
me the course and I will confirm."),
    ("timing", ["timing", "timings", "schedule", "when"], "Classes run on weekends. Which
batch do you prefer?"),
```

```
    ("booking", ["book", "enrol", "enroll", "join", "seat"], "Great! Please share your name
and phone number and a counsellor will call you."),
]
HANDOFF = "I will connect you with a counsellor for that."

def normalise(text):
    return re.sub(r"[^a-z ]", " ", text.lower())

def reply(message):
    words = normalise(message)
    for name, keywords, answer in INTENTS:
        for kw in keywords:
            if re.search(rf"\b{re.escape(kw)}\b", words):
                return name, answer
    return "handoff", HANDOFF

if __name__ == "__main__":
    assert reply("Hi there")[0] == "greeting"
    assert reply("how much is the course?")[0] == "price"
    assert reply("Can I book a seat for Monday?")[0] == "booking"
    assert reply("When are the timings?")[0] == "timing"
    assert reply("Is there a refund policy?")[0] == "handoff"
    assert reply("this is fine")[0] == "handoff"   # 'hi' inside 'this' must not match
    print("chatbot tests passed")
```

Assignment with full solution

["Matching partial words, so 'this' triggers the greeting.", 'No handover, so questions are lost.', 'Promising prices or dates the bot cannot verify.', 'No monthly test set, so quality drifts unnoticed.']

Common mistakes

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Freelancing opportunities

Worked example. A visitor asks 'how much is the course?'. The bot matches the word 'how much', returns the price answer, and asks which course. If nothing matches, it hands over with the captured question.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Botpress documentation
- Voiceflow documentation
- WhatsApp Business developer documentation

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 4

Machine learning engineer

Profession 4 of 10 in Development and IT.

What the profession means

A machine learning engineer builds systems that learn patterns from data and make predictions: lead scores, demand forecasts, fraud flags. The work covers data preparation, model training and testing, deployment and monitoring.

What professionals do every day

Cleans and prepares data, selects features, trains and compares models, measures them honestly on data they have not seen, deploys the chosen model and watches it for drift.

Why businesses hire them, and what problems they solve

Many business decisions depend on predictions: who will buy, what will sell, which invoice is risky. An engineer turns historical data into a model the business can use, and says how much to trust it.

- Guessing which leads to call first.
- Stock or demand planning based on gut feel.
- Manual checks on large volumes of transactions.

Skills you need

- Python, NumPy and pandas.
- Statistics: averages, spread, correlation, the idea of a test set.
- Scikit-learn for classical models.
- Evaluation: accuracy, precision, recall and the cost of each error.
- Deployment basics and monitoring.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python	Language for data and ML	Free	Free	www.python.org
scikit-learn	Classical ML models	Free	Free	scikit-learn.org
pandas	Data tables	Free	Free	pandas.pydata.org
Jupyter	Notebooks for experiments	Free	Free	jupyter.org
Kaggle	Datasets and practice competitions	Free	Paid features	www.kaggle.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Clean a dataset and train one classifier with a test set.
Intermediate	Compare three models and explain the choice.

Advanced	Deploy a model behind an API and log its predictions.

Projects by level

- **Beginner:** Lead scoring for a sales team.
- **Intermediate:** Churn prediction with a retention plan.
- **Advanced:** Demand forecast with error bands.

Client-ready portfolio project

A lead-scoring model for a fictional education company: data preparation, a model, an honest test score, and a one-page note on what the score can and cannot tell the sales team.

Worked example

ASSIGNMENT

A lead-scoring model with an honest test score

Business problem: A sales team wants to call the most likely buyers first. The model must be tested on data it did not learn from.

Steps

1. Load the data and check for missing values.
2. Split into training and test sets with a fixed random seed.
3. Train a logistic regression model.
4. Report accuracy on the test set and state the expected error cost (for example a missed buyer).
5. Write one paragraph on what the score does not show.

Expected output: A test accuracy above the agreed threshold (here 0.8) and a written limitation note.

SOLUTION (ANSWER KEY)

The test set is held back during training, so the score reflects new data. A fixed `random_state` makes the result repeatable. Accuracy alone can hide problems when one outcome is rare; check precision and recall in a real project. The test here passes at about 0.86 accuracy on synthetic data.

PYTHON

```
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

rng = np.random.default_rng(42)
n = 400
time_on_site = rng.uniform(0, 10, n)          # minutes on the pricing page
emails_opened = rng.integers(0, 8, n)         # emails opened this month
noise = rng.normal(0, 1.0, n)
score = 0.6 * time_on_site + 0.5 * emails_opened + noise
converted = (score > 5.5).astype(int)         # 1 = became a paying customer

X = np.column_stack([time_on_site, emails_opened])
```

```
X_train, X_test, y_train, y_test = train_test_split(X, converted, test_size=0.25,
random_state=0)
model = LogisticRegression().fit(X_train, y_train)
acc = accuracy_score(y_test, model.predict(X_test))
if __name__ == "__main__":
    print("test accuracy:", round(acc, 3))
    assert acc > 0.8, acc
    print("lead model test passed")
```

Assignment with full solution

['Testing on the training data, which makes results look better than they are.', 'Using a model without checking for data leakage (information from the future).', 'Reporting only accuracy when classes are unbalanced.', 'Not monitoring a deployed model as customer behaviour changes.']

Common mistakes

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Freelancing opportunities

Worked example. Time on the pricing page and emails opened predict conversion. The model is trained on 75% of the data and scored on the remaining 25%, which it has never seen.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- scikit-learn user guide
- pandas documentation
- Kaggle Learn (free micro-courses)

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 5

AI developer

Profession 5 of 10 in Development and IT.

What the profession means

An AI developer builds applications that use AI models at their core: assistants, document tools, generators and agents. The developer writes the application code, the model calls, the data handling and the user interface.

What professionals do every day

Designs the feature, writes prompts and evaluation sets, builds the application and API layer, adds retrieval where documents matter, tests quality and deploys with monitoring and cost limits.

Why businesses hire them, and what problems they solve

Businesses want AI features inside their products without building models from scratch. An AI developer delivers these features reliably, safely and at a predictable cost.

- Feature ideas that need a working prototype.
- Document-heavy work that is slow to read and answer.
- Customer-facing AI that must be accurate and safe.

Skills you need

- Python or JavaScript application development.
- Model APIs, structured outputs and function calling.
- RAG and vector search (see Module D).
- Evaluation: test sets, scoring and regression checks.
- Security, privacy and cost monitoring.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
OpenAI API	Models and tools	Pay per use	Pay per use	platform.openai.com/docs
Anthropic API	Claude models and tools	Pay per use	Pay per use	docs.claude.com
Google AI Studio	Gemini models	Free tier	Pay per use	aistudio.google.com
Cursor	Code editor with AI	Free tier	Paid plans	cursor.com
Vercel	Hosting for AI apps	Hobby tier	Paid plans	vercel.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	A single-page app that calls a model and shows the result.
Intermediate	A document question-answer app with sources.

Advanced	An app with login, usage limits and an evaluation report.

Projects by level

- **Beginner:** Summary tool for meeting notes.
- **Intermediate:** Q&A over a company handbook.
- **Advanced:** Proposal writer with a review step and history.

Client-ready portfolio project

A document assistant for a fictional firm: upload or paste documents, ask questions, get sourced answers, and see an evaluation report on a test set.

Worked example

ASSIGNMENT

Evaluate an AI feature before release

Business problem: A team wants to ship a document Q&A feature. They need evidence it answers correctly, and they must know when it fails.

Steps

1. Write 20 test questions with the correct answer and the source page.
2. Run the feature on each question and record whether the answer is correct and cited.
3. Count the failures and group them by cause (retrieval miss, wrong fact, no answer).
4. Fix one cause and rerun the full set to check nothing else broke.

Expected output: A table of results with a pass rate, failure causes and a before-and-after comparison.

SOLUTION (ANSWER KEY)

Evaluation makes quality visible. Retrieval misses and wrong facts need different fixes. Rerunning the whole set after each change catches regressions. The RAG example in Module D is a tiny version of this loop.

PYTHON

```
import re
from collections import Counter

STOP = {"the", "and", "for", "are", "you", "your", "how", "what", "does", "long", "take",
        "have", "with", "from", "that", "this", "after", "within", "each", "shown", "before",
"run", "our", "can", "will"}
DOCS = {
    "refund": "Refunds are processed within 7 working days after a written request.",
    "fees": "Course fees are shown on each program page and confirmed in writing before
payment.",
    "timings": "Live classes run on Saturdays and Sundays in the evening.",
}

def tokens(text):
    return [w for w in re.findall(r"[a-z]+", text.lower()) if len(w) > 2 and w not in STOP]
```

```
def retrieve(question, docs=DOCS, k=1):
    q = Counter(tokens(question))
    scored = []
    for key, text in docs.items():
        d = Counter(tokens(text))
        score = sum(min(q[w], d[w]) for w in q)
        scored.append((score, key, text))
    scored.sort(reverse=True)
    return [(key, text) for score, key, text in scored[:k] if score > 0]

def answer(question):
    hits = retrieve(question)
    if not hits:
        return "Not found in our documents. A team member will reply."
    key, text = hits[0]
    return f"{text} [source: {key}]"

if __name__ == "__main__":
    assert answer("How long do refunds take?").endswith("[source: refund]")
    assert "7 working days" in answer("How long do refunds take?")
    assert answer("What are the fees for the course?").endswith("[source: fees]")
    assert answer("When do the live classes run?").endswith("[source: timings]")
    assert answer("Do you sell pizza?").startswith("Not found")
    print("rag tests passed")
```

Assignment with full solution

['Shipping without an evaluation set.', 'Trusting one good demo.', "Sending whole private documents to a model without checking the provider's data terms.", 'Ignoring cost per request.']

Common mistakes

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Freelancing opportunities

Worked example. A question goes to a retriever that returns the best passages; the model is asked to answer only from them and cite the page. The app shows the passages next to the answer so the user can check.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- OpenAI API documentation
- Claude documentation
- Google AI for developers

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 6

Automation expert

Profession 6 of 10 in Development and IT.

What the profession means

An automation expert designs and runs workflows that move data and trigger actions between business tools, using platforms such as n8n, Make, Zapier or custom scripts. They look at a process end to end and remove the manual steps.

What professionals do every day

Maps the process, chooses the tool, builds and tests the workflow, adds error alerts and approval steps, documents it and maintains it as the business changes.

Why businesses hire them, and what problems they solve

Manual copy-paste between systems costs time and causes errors. An automation expert connects the systems so the data is correct and the team's time goes to work that needs judgement.

- Copying leads from forms into spreadsheets and CRMs by hand.
- Reports assembled manually every week.
- Missed follow-ups.

Skills you need

- Process mapping and choosing the right trigger.
- Triggers, actions, filters, data mapping.
- Error handling, retries and alerts.
- Credentials and security.
- Documentation and handover.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
n8n	Node-based workflows, self-hostable	Self-hosted free	Cloud plans	n8n.io
Make	Visual scenarios	Free tier	Paid plans	www.make.com
Zapier	Simple linear automations	Free tier	Paid plans	zapier.com
Google Sheets	Data store for many flows	Free	Paid Workspace plans	sheets.google.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Connect a form to a spreadsheet and send a confirmation.

Intermediate	Route leads by source with an alert.
Advanced	Weekly report generated and sent automatically.

Projects by level

- **Beginner:** Lead capture to CRM with follow-up.
- **Intermediate:** Invoice reminders with approval.
- **Advanced:** Monthly business dashboard feed.

Client-ready portfolio project

A lead-to-follow-up automation for a fictional clinic, with an approval step, error alerts, a test log and a one-page handover.

Worked example

ASSIGNMENT

Lead automation with an approval step

Business problem: A clinic wants every enquiry saved, answered within an hour, and followed up once. Nothing should be sent to a patient without a staff check.

Steps

1. Map the process on paper: trigger, save, draft reply, approval, send, follow-up.
2. Build the flow in n8n or Make with a test form.
3. Add an error alert to a staff channel.
4. Test with three records: a complete one, one with a missing phone number, and one duplicate.
5. Write the handover: what runs, what it costs, and how to pause it.

Expected output: A working flow, three test results and a handover page.

SOLUTION (ANSWER KEY)

The approval step is the safety net: no message leaves without a person. The missing-phone case must fail visibly, not silently. Duplicate detection prevents a patient receiving two replies. The n8n guide in the AI tools series shows a complete Gmail example with the same structure.

Assignment with full solution

['Switching on an automation before testing with sample data.', 'No error alerts, so failures stay hidden.', 'Hard-coding secrets into the workflow.', 'No handover, so nobody can maintain it later.']

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Freelancing opportunities

Worked example. Form submitted, lead row added, a message sent after an approval, and an alert if any step fails. Each step is tested with one real-looking record before it is switched on.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- n8n documentation
- Make help centre
- Zapier help centre

n8n expert

Profession 7 of 10 in Development and IT.

What the profession means

An n8n expert builds advanced workflows in n8n: nodes, expressions, error handling, sub-workflows, AI nodes and self-hosting. n8n is popular with developers and operations teams because workflows can include custom code and run on the company's own server.

What professionals do every day

Designs workflows, writes expressions and small code steps, handles errors and retries, connects databases and APIs, sets up credentials safely, self-hosts when required and monitors executions.

Why businesses hire them, and what problems they solve

Companies with many systems need reliable, customisable automation. An n8n expert delivers it with control over data and cost.

- Complex processes that simple tools cannot handle.
- Data that must stay on the company's own server.
- Failed workflows that nobody notices.

Skills you need

- n8n nodes, triggers, connections and expressions.
- JSON and webhooks (Development and IT, Part 1).
- Error workflows and monitoring.
- Credentials and secret handling.
- Self-hosting with Docker (official n8n guide).

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
n8n	Workflow automation with code steps	Community self-hosted	Cloud plans	n8n.io
n8n documentation	Official reference	Free	Free	docs.n8n.io
n8n learning portal	Official courses	Free	Free	learn.n8n.io
Docker	Run n8n in a container	Free	Free	www.docker.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish

Beginner	A workflow with a trigger, two nodes and a condition.
Intermediate	A workflow with an error handler and an alert.
Advanced	A sub-workflow reused by three parent flows, with monitoring.

Projects by level

- **Beginner:** Lead capture with CRM and email follow-up.
- **Intermediate:** Customer support triage with AI classification.
- **Advanced:** Weekly business report from three data sources.

Client-ready portfolio project

A complete n8n system for a fictional business: lead capture, AI triage, CRM update, error alerting and a monitoring page. See the n8n complete guide in the AI tools series for the Gmail version.

Worked example

ASSIGNMENT

Error handling for a production workflow

Business problem: A workflow sends customer emails. If the email service fails, the team must know, and the lead must not be lost.

Steps

1. Add a Set node to keep only the fields you need.
2. Save each lead to a sheet before sending any email.
3. Add an error workflow that sends a staff alert with the execution link.
4. Break the email credential on purpose and confirm the alert arrives.
5. Restore the credential and confirm the lead is still saved and the email sends once.

Expected output: A failing test that triggers an alert, and proof that no lead was lost.

SOLUTION (ANSWER KEY)

Saving first and sending second means a failed email never loses a lead. The error workflow converts silent failures into alerts. Deliberately breaking a credential in a test is the only reliable way to prove the alert works.

Assignment with full solution

['Storing secrets in expressions instead of credentials.', 'No error workflow, so failures go unnoticed.', 'Running a self-hosted instance without updates and backups.', 'Large loops without limits, which can exhaust API quotas.']

Common mistakes

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Freelancing opportunities

Worked example. A webhook receives a lead. A condition routes hot leads to a call task and others to a nurture email. An error workflow sends an alert whenever any run fails.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- n8n documentation
- n8n learning portal

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 8

Vibe coder

Profession 8 of 10 in Development and IT.

What the profession means

A vibe coder builds working software by describing what they want in plain language to an AI coding tool, then reviewing, testing and steering the result. The skill is in the specification, the review and the judgement, not in typing code from scratch.

What professionals do every day

Writes clear product requirements as prompts, reviews generated code, runs the app, reproduces and fixes errors, connects a database and authentication, deploys and checks security before launch.

Why businesses hire them, and what problems they solve

Small businesses and founders need working prototypes and simple internal tools quickly. A vibe coder delivers them in days, and knows when to bring in a developer.

- Ideas that need a working prototype to test with customers.
- Internal tools that replace spreadsheets.
- Landing pages and simple web apps with a database.

Skills you need

- Writing a product spec: users, data, screens, rules.
- Reading generated code well enough to spot problems.
- Debugging from error messages.
- Database and authentication basics; security review.
- Deployment and when to hand over to a developer.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Lovable	Full-stack apps from a conversation	Free tier	Paid plans	lovable.dev
Bolt	Prompt-to-app builder	Free tier	Paid plans	bolt.new
Replit	Browser environment with an AI agent	Free tier	Paid plans	replit.com
Cursor	AI code editor	Free tier	Paid plans	cursor.com
Claude Code	Coding agent in the terminal	Included with some Claude plans	Pay per use or plan	docs.claude.com/en/docs/claude-code/overview
GitHub Copilot	AI pair programmer	Free tier for some users	Paid plans	github.com/features/copilot

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	A one-screen tool built from a written spec.
Intermediate	A form that saves to a database with validation.
Advanced	A multi-user app with login and a role check, tested with two accounts.

Projects by level

- **Beginner:** Booking page for a clinic.
- **Intermediate:** Job application tracker with reminders.
- **Advanced:** Client portal with files and invoices (with a developer review before launch).

Client-ready portfolio project

A business web app (booking, tracker or simple SaaS) built from a written spec, with two-account security testing, a README and a handover note.

Worked example

ASSIGNMENT

Spec, build and secure a simple app

Business problem: A job seeker wants a tracker where they can see their applications by status. Only they can see their own data.

Steps

1. Write a one-page spec: users, fields, statuses, rules.
2. Build the board screen first and test it on a phone.
3. Add the database with row-level security so each user sees only their rows.
4. Create two accounts and try to read the other account's rows through the app and the address bar.
5. Write a README and a list of what a developer should review before a public launch.

Expected output: A working app, a passing two-account test and a review checklist.

SOLUTION (ANSWER KEY)

The two-account test is the most important step. If account A can read account B's data through any route, the app is not ready. The spec makes the review possible: you check the app against the rules you wrote.

Assignment with full solution

['Accepting generated code you have not read.', "Skipping security checks because the app 'works'.", 'Writing prompts without a spec, which produces inconsistent screens.', 'Launching with secret keys in

the front-end code.']

Common mistakes

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Freelancing opportunities

Worked example. A spec for a job tracker: users, applications, statuses, reminders. The prompt asks for one screen first, then the database, then validation, then reminders. Each step is tested before the next.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Lovable documentation
- Replit documentation
- Supabase: row-level security guide

Cloud expert

Profession 9 of 10 in Development and IT.

What the profession means

A cloud expert sets up and runs computing, storage, databases and AI services on cloud platforms such as AWS, Microsoft Azure and Google Cloud. The work covers accounts and access, hosting, networks, cost control, security and monitoring.

What professionals do every day

Designs the architecture, provisions services, sets permissions with least privilege, deploys applications, monitors costs and performance, and plans backups and recovery.

Why businesses hire them, and what problems they solve

Applications need reliable hosting, secure access and costs that stay predictable. A cloud expert makes the platform run well and shows where the money goes.

- Servers that go down or cost more than planned.
- Access that is too broad and a security risk.
- Data that is not backed up or cannot be restored.

Skills you need

- Cloud basics: regions, compute, storage, databases, networks.
- Identity and access management (least privilege).
- Infrastructure basics and deployment.
- Cost management and budgets.
- Monitoring, backups and disaster recovery.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
AWS	Broad cloud platform	Free tier for many services	Pay per use	aws.amazon.com/free
Microsoft Azure	Cloud and AI services	Free account credits	Pay per use	azure.microsoft.com/free
Google Cloud	Cloud and AI services	Free tier	Pay per use	cloud.google.com/free
Vercel	Hosting for web apps	Hobby tier	Paid plans	vercel.com
Netlify	Hosting for static sites and functions	Free tier	Paid plans	www.netlify.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Host a static site on a free tier with HTTPS.
Intermediate	Set a budget alert and a spending limit.
Advanced	Create a database with backups and test one restore.

Projects by level

- **Beginner:** Deployment pipeline for a web app.
- **Intermediate:** Cost report and optimisation for a client.
- **Advanced:** Backup and recovery plan with a tested restore.

Client-ready portfolio project

A hosted application for a fictional business with least-privilege access, a budget alert, automated backups, one tested restore and a one-page cost summary.

Worked example

ASSIGNMENT

Budget alert and a tested restore

Business problem: A client's app runs on a cloud account. The client fears unexpected bills and data loss.

Steps

1. Create a budget with an alert at 50% and 80% of the monthly limit (set it in the provider's billing console).
2. Create a role for the application with access to one database only.
3. Turn on automated backups and record the schedule.
4. Restore a backup to a test copy and check three records match.
5. Write a one-page summary: limits, owners, backup schedule and the restore steps.

Expected output: Alerts configured, a restore test log with matching records, and a one-page summary.

SOLUTION (ANSWER KEY)

Alerts and least-privilege roles limit both cost and damage. A backup you have never restored is a hope, not a plan. Exact console steps change; follow the provider's current documentation.

Assignment with full solution

['Using the root or admin account for everyday tasks.', 'Leaving test resources running and paying for them.', 'Storing backups in the same place as the live data.', 'Never testing a restore.']

Common mistakes

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Freelancing opportunities

Worked example. A small app needs a database and a web server. The cloud expert gives the app its own role with permission only for that database, sets a budget alert at a fixed amount, and schedules a backup. The restore is tested, not assumed.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- AWS Training (official)
- Microsoft Learn: Azure
- Google Cloud Skills (official)

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 10

AI consultant

Profession 10 of 10 in Development and IT.

What the profession means

An AI consultant helps a business decide where AI will help, how much it will return and how to implement it safely. The consultant audits processes, proposes solutions, calculates the return, and supports implementation and training.

What professionals do every day

Interviews staff, maps processes, finds opportunities, estimates benefit and cost, writes proposals, runs a proof of concept, supervises implementation and trains the team.

Why businesses hire them, and what problems they solve

Many businesses are unsure where to start and afraid of wasting money. A consultant gives a clear, costed plan with a small first step.

- Time lost on repetitive work.
- Unclear priorities and expensive tool purchases.
- Risk: data, accuracy and staff concerns.

Skills you need

- Process mapping and opportunity scoring.
- ROI and payback calculations (below).
- Proposal writing and client communication.
- AI risks: accuracy, privacy, bias, change management.
- Training and handover.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Microsoft Excel	Models and ROI tables	Free online version	Paid plans	www.microsoft.com/microsoft-365/excel
Google Sheets	Shared models	Free	Paid Workspace plans	sheets.google.com
Miro	Process maps and workshops	Free tier	Paid plans	miro.com
n8n	Prototype automations	Self-hosted free	Cloud plans	n8n.io

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish

Beginner	Estimate the saving for one process.
Intermediate	Write a one-page opportunity summary.
Advanced	Run a two-week proof of concept with clear success criteria.

Projects by level

- **Beginner:** Real estate: automate enquiry handling and site visit booking.
- **Intermediate:** Education: automate course enquiries and attendance reminders.
- **Advanced:** Retail: automate stock alerts and supplier orders.

Client-ready portfolio project

A complete AI automation proposal for a fictional real estate company: process map, opportunity list, ROI calculation, risks, a two-week pilot plan and a training outline.

Worked example

ASSIGNMENT

Estimate the return on an automation

Business problem: A firm asks whether automating enquiry replies will pay for itself.

Steps

1. List the hours saved per week, the hourly cost, and the working weeks per year.
2. Enter the build cost and the monthly tool cost.
3. Compute the first-year net, the payback period and the return.
4. State the assumptions and what would change the answer.
5. Show the result as a range (low and high hours saved), not one number.

Expected output: A table of results with assumptions, a payback period and a range.

SOLUTION (ANSWER KEY)

Every figure is an assumption until measured. The calculator returns the same numbers as the worked example: a saving of Rs 96,000, a first-year net of Rs 12,000, payback in 10 months, and a return of 14.3%. Present a range, because the hours saved are an estimate.

PYTHON

```
def automation_roi(hours_saved_per_week, hourly_cost, weeks_per_year, build_cost,
monthly_tool_cost):
    annual_saving = hours_saved_per_week * hourly_cost * weeks_per_year
    annual_tool = monthly_tool_cost * 12
    first_year_net = annual_saving - build_cost - annual_tool
    net_per_year_after_build = annual_saving - annual_tool
    payback_months = build_cost / (net_per_year_after_build / 12) if net_per_year_after_build
> 0 else float("inf")
    roi_percent = first_year_net / (build_cost + annual_tool) * 100
    return {"annual_saving": annual_saving, "annual_tool": annual_tool,
            "first_year_net": first_year_net, "payback_months": round(payback_months, 1),
            "roi_percent": round(roi_percent, 1)}
```

```
if __name__ == "__main__":
    r = automation_roi(hours_saved_per_week=5, hourly_cost=400, weeks_per_year=48,
                       build_cost=60000, monthly_tool_cost=2000)
    assert r["annual_saving"] == 96000 and r["annual_tool"] == 24000
    assert r["first_year_net"] == 12000
    assert r["payback_months"] == 10.0
    assert r["roi_percent"] == 14.3
    print("roi tests passed:", r)
```

Assignment with full solution

['Promising a return without measuring the current process.', 'Ignoring the cost of staff time during implementation.', 'No pilot, so the full project starts before anyone has tested it.', 'Ignoring privacy and data-handling rules.']

Common mistakes

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Freelancing opportunities

Worked example. Staff spend 5 hours a week copying enquiries. At Rs 400 an hour over 48 working weeks, the saving is Rs 96,000 a year. Subtract a tool cost of Rs 24,000 a year and the build cost gives the payback period and return. The calculator in Module H shows the arithmetic.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- AI for Everyone (Coursera, check current availability and fees)
- NIST AI Risk Management Framework

MODULE C

AI video creation and editing: process, tools and quality

Short video is how most businesses now explain products. This module covers the full production path.

C1. Types of video work

Type	Typical use	Common mistake
Text-to-video and image-to-video	Ad concepts, B-roll, product scenes	Unclear motion, inconsistent characters between clips
AI avatars and presenters	Explainers, training, announcements	Using a real person likeness without written permission
Voiceovers, subtitles, translation and dubbing	Multilingual ads, accessibility	Not checking names and numbers in generated subtitles
Background removal and editing	Product videos, reels, cut-downs	Visible edges and flicker around the subject
Social reels and YouTube production	Regular content for brands	No hook in the first three seconds

C2. Product and advertising videos

A product video answers three questions in order: what it is, why it matters to this viewer, and what to do next. Write the three answers before opening a tool.

C3. Educational videos

One idea per video, a worked example on screen, and a short recap. Add captions to every video.

C4. Production checklist

1. Script and storyboard approved.
2. Prompt and settings logged for each clip.
3. Voice and subtitles checked against the script.
4. Brand colours, logo and fonts licensed for use.
5. Exported in each platform size; file names and captions delivered with the video.

VERIFICATION NOTE

Tool names, features, plan names and prices change often. The links below point to official sites but were not checked live in this session. Confirm each one before you rely on it or quote it to a client.

MODULE D

AI integration and chatbot projects

Three complete chatbot-family projects, from the simplest to a document assistant. Each has a test log.

Project 1: FAQ bot for a coaching centre

Four intents (greeting, price, timing, booking), a fallback handover and a test set of ten messages. The complete code is in the Chatbot developer chapter; run its tests before you publish.

PYTHON

```
import re

INTENTS = [
    ("greeting", ["hi", "hello", "hey"], "Hello! I can help with prices, timings and bookings."),
    ("price", ["price", "cost", "fee", "fees", "how much"], "Our fees start at Rs 5,000. Tell me
the course and I will confirm."),
    ("timing", ["timing", "timings", "schedule", "when"], "Classes run on weekends. Which batch do
you prefer?"),
    ("booking", ["book", "enrol", "enroll", "join", "seat"], "Great! Please share your name and
phone number and a counsellor will call you."),
]
HANDOFF = "I will connect you with a counsellor for that."

def normalise(text):
    return re.sub(r"[^a-z ]", " ", text.lower())

def reply(message):
    words = normalise(message)
    for name, keywords, answer in INTENTS:
        for kw in keywords:
            if re.search(rf"\b{re.escape(kw)}\b", words):
                return name, answer
    return "handoff", HANDOFF

if __name__ == "__main__":
    assert reply("Hi there")[0] == "greeting"
    assert reply("how much is the course?")[0] == "price"
    assert reply("Can I book a seat for Monday?")[0] == "booking"
    assert reply("When are the timings?")[0] == "timing"
    assert reply("Is there a refund policy?")[0] == "handoff"
    assert reply("this is fine")[0] == "handoff"   # 'hi' inside 'this' must not match
    print("chatbot tests passed")
```

Project 2: lead-capture bot with a CRM

1. Greeting and two questions: course and phone number.
2. Validate the phone number on the server (10 digits after removing symbols).
3. Create a lead in the CRM through its API or an automation tool.
4. Send a template confirmation within the messaging platform's rules.
5. Test with a valid lead, an invalid phone number and a duplicate.

Project 3: knowledge assistant over documents

Retrieval first, then the model answers only from the retrieved passages and cites them. The small retrieval program below shows the principle without any API key.

PYTHON

```
import re
from collections import Counter

STOP = {"the", "and", "for", "are", "you", "your", "how", "what", "does", "long", "take",
        "have", "with", "from", "that", "this", "after", "within", "each", "shown", "before",
"run", "our", "can", "will"}
DOCS = {
    "refund": "Refunds are processed within 7 working days after a written request.",
    "fees": "Course fees are shown on each program page and confirmed in writing before payment.",
    "timings": "Live classes run on Saturdays and Sundays in the evening.",
}

def tokens(text):
    return [w for w in re.findall(r"[a-z]+", text.lower()) if len(w) > 2 and w not in STOP]

def retrieve(question, docs=DOCS, k=1):
    q = Counter(tokens(question))
    scored = []
    for key, text in docs.items():
        d = Counter(tokens(text))
        score = sum(min(q[w], d[w]) for w in q)
        scored.append((score, key, text))
    scored.sort(reverse=True)
    return [(key, text) for score, key, text in scored[:k] if score > 0]

def answer(question):
    hits = retrieve(question)
    if not hits:
        return "Not found in our documents. A team member will reply."
    key, text = hits[0]
    return f"{text} [source: {key}]"

if __name__ == "__main__":
    assert answer("How long do refunds take?").endswith("[source: refund]")
    assert "7 working days" in answer("How long do refunds take?")
    assert answer("What are the fees for the course?").endswith("[source: fees]")
    assert answer("When do the live classes run?").endswith("[source: timings]")
    assert answer("Do you sell pizza?").startswith("Not found")
    print("rag tests passed")
```

Security and privacy for all three

- No personal data in logs.
- Clear notice that the visitor is talking to a bot, and how to reach a person.
- Opt-out and data deletion on request.

MODULE E

n8n and workflow automation: eight mandatory projects

Each project follows the same pattern: trigger, save, process, approve, send, log, test. The full Gmail example is in the n8n complete guide.

#	Project	Trigger	Key steps	Test
1	Lead capture	Form webhook	Validate, save to sheet, confirm	Valid, invalid, duplicate
2	Website enquiry automation	Form webhook	Classify enquiry, route to team	Each category
3	Email automation	New email label	Classify, draft reply, label, approve	Sample emails, no auto-send
4	Google Sheets automation	Row added	Calculate, format, notify	Empty cells, text in number columns
5	CRM lead management	Lead saved	Assign owner, set follow-up date	Round-robin assignment
6	Appointment booking	Booking request	Check slot, confirm, reminder	Full slot, free slot
7	AI customer support	Support message	Classify, answer from documents, hand over	Known answer, unknown answer
8	Automated business reporting	Weekly schedule	Collect data, compute totals, send summary	Totals match source

How to test any project

1. Use test data only, and a test recipient.
2. Check every node's output for one record.
3. Run the three test cases in the table.
4. Turn on error alerts before going live.

MODULE F

Vibe coding: prompts, review, debugging and launch

Vibe coding means building software by describing it in plain language, then reviewing and testing the result. The tools are listed in the AI tools directory (Lovable, Bolt, Replit, Cursor, Claude Code, GitHub Copilot).

F1. Writing a development prompt

```
Build a booking page for a clinic.
Users: patients (no login) and staff (login).
Screens: service picker, date and time, confirmation.
Data: bookings with name, phone, service, time.
Rules: no double bookings; phone must be 10 digits.
First build the patient screen only. Explain each file you change.
```

F2. Review checklist for generated code

- Does every rule in the spec appear in the code?
- Are secrets kept out of front-end files?
- Are inputs validated on the server?
- Do permissions stop one user seeing another's data?
- Are error messages useful and safe?

F3. Debugging

1. Copy the exact error message into the chat.
2. Ask what the error means before asking for a fix.
3. Change one thing and test again.

F4. Launch

Before a public launch, run the security review in the Development and IT textbook (Part 3, Module D) and test with two accounts.

MODULE G

Machine learning and AI engineering: a learning roadmap

A staged path from Python to deployment. Each stage ends with a small project.

Stage	Learn	Project
1. Python	Variables, lists, functions, files	Clean a CSV file
2. Maths and statistics	Averages, spread, probability, correlation	Describe a sales dataset
3. NumPy and pandas	Arrays, tables, grouping, missing values	Build a customer summary
4. Scikit-learn	Train, test, metrics, feature engineering	Lead-scoring model (see the ML engineer chapter)
5. Deep learning (PyTorch or TensorFlow)	Neural networks, training loops	Image classifier on a public dataset
6. NLP and computer vision	Text classification, image tasks	Ticket classifier
7. Deployment and MLOps	APIs, monitoring, retraining	Model behind an API with logging

Official resources to check: scikit-learn, PyTorch tutorials, TensorFlow tutorials, Kaggle Learn. Links not checked live.

MODULE H

AI consulting: audits, ROI, proposals and case studies

How consultants find the first AI opportunity, price it and deliver it. Case studies cover real estate, education, healthcare, retail and e-commerce.

H1. Audit steps

1. List the processes that repeat every day or week.
2. Score each by hours spent, error cost and data readiness.
3. Pick one process for a two-week proof of concept.
4. Define success in numbers before starting.

H2. ROI calculation

The calculator below is tested. Present the result as a range, because hours saved are estimates.

PYTHON

```
def automation_roi(hours_saved_per_week, hourly_cost, weeks_per_year, build_cost,
monthly_tool_cost):
    annual_saving = hours_saved_per_week * hourly_cost * weeks_per_year
    annual_tool = monthly_tool_cost * 12
    first_year_net = annual_saving - build_cost - annual_tool
    net_per_year_after_build = annual_saving - annual_tool
    payback_months = build_cost / (net_per_year_after_build / 12) if net_per_year_after_build > 0
else float("inf")
    roi_percent = first_year_net / (build_cost + annual_tool) * 100
    return {"annual_saving": annual_saving, "annual_tool": annual_tool,
            "first_year_net": first_year_net, "payback_months": round(payback_months, 1),
            "roi_percent": round(roi_percent, 1)}

if __name__ == "__main__":
    r = automation_roi(hours_saved_per_week=5, hourly_cost=400, weeks_per_year=48,
                       build_cost=60000, monthly_tool_cost=2000)
    assert r["annual_saving"] == 96000 and r["annual_tool"] == 24000
    assert r["first_year_net"] == 12000
    assert r["payback_months"] == 10.0
    assert r["roi_percent"] == 14.3
    print("roi tests passed:", r)
```

H3. Case studies

Sector	Opportunity	Measure
Real estate	Enquiry replies, site-visit booking, document checks	Hours per week; days to first reply
Education	Course enquiries, fee reminders, attendance messages	Enquiry conversion; late payments
Healthcare (clinics)	Appointment reminders, intake forms, FAQ bot (no clinical advice)	No-show rate; front-desk time
Retail	Stock alerts, supplier orders, returns triage	Stock-outs; time per order

E-commerce	Order sync, review replies, abandoned-cart messages (with consent)	Order errors; response time

H4. Mandatory assignment: AI automation proposal for a fictional real estate company

1. Process map: enquiry in, qualification, site visit booked, follow-up, paperwork.
2. Opportunities: auto-reply with qualification questions; booking link; document checklist reminders.
3. ROI: 6 hours saved per week at Rs 500 an hour over 48 weeks, tool cost Rs 3,000 a month, build cost Rs 80,000 (illustrative; run the calculator with your figures).
4. Risks: accuracy of replies, personal data, staff adoption, regulatory notices.
5. Pilot: two weeks, one property type, success measured against the baseline.
6. Training: 60 minutes for staff, with a one-page guide and an escalation route.

Solution note. Use the calculator: 6 × 500 × 48 = Rs 144,000 gross saving; minus Rs 36,000 tool cost a year and Rs 80,000 build cost gives a first-year net of Rs 28,000 (before any other costs). Present this with its assumptions.

VERIFICATION NOTE

Tool names, features, plan names and prices change often. The links below point to official sites but were not checked live in this session. Confirm each one before you rely on it or quote it to a client.

EMERGING PROFESSIONS

Emerging AI roles: pathways built from the core chapters

These roles are built from the chapters above, not taught as separate subjects. Demand for each should be re-checked before you promote it.

Profession	Pathway (start here)	Portfolio project
AI agent developer	Module D chatbot projects, then agents (see the AI agents guide)	Agent with approval gates and tests
RAG developer	Module D project 3, then vector search (see the RAG guide)	Document Q&A with evaluation set
LLM application developer	AI integration and AI developer chapters	Application with validation and cost control
AI workflow architect	n8n expert and automation expert chapters	Architecture diagram and tested flow
AI evaluation specialist	AI developer chapter (evaluation assignment)	Test set, scores and failure report
AI voice-agent developer	Chatbot developer chapter plus speech tools (verify)	Voice flow with human handover
AI customer-support automation specialist	Chatbot and automation chapters	Triage and reply system with escalation
AI sales-automation specialist	Automation expert chapter plus CRM basics	Lead routing and follow-up sequence
AI research-automation specialist	RAG and automation chapters	Source-cited research digest
AI data-automation specialist	ML and automation chapters	Scheduled data pipeline with checks